Demo Paper Title

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***Abstract*—Write a short abstract about your demo paper: What will you show and why is this important?**

***Index Terms*—Insert a comma-separated list of keywords describing your demo, e.g. forecasting, time series, the name of your method,...**

I. Introduction

Write one or two paragraphs where you describe the problem you try to solve, explain why this is a problem, e.g., planning for purchases of ingredients, preservation of the shop's offer, or a more general explanation why forecasting makes sense in many scenarios. In the latter case, the coffee sales just act as an example. Give a short pitch of how you treat the challenge.

II. System/Solution/Architecture Overview

Provide an overview of your solution. Use the figure you created for the last exercise sheet to explain your solution. You might adjust your figure to fit into the layout and to stay within the page limit. Include a short description of the forecasting approach you are using, ideally also providing a source.

III. Demonstration Proposal/Scenarios/Workflow

This section describes how users can experience our forecasting pipeline. User can fine-tune the model and see where the model behaves well as well as where its assumptions start to break down or find them automatically via a grid search

A. Setup

The demo runs on a laptop with an Apple Silicon M3 CPU and 16GB of RAM on macOS, however, the demo can also be run on less powerful hardware, given the moderate size of the dataset. The forecasting core is implemented in Java 17 and built with Maven into a self-contained jar. Historical sales are stored in PostgreSQL 16, while the ingredient recipes used to translate drink forecasts into ingredient quantities are kept in a hosted MongoDB Atlas cluster. Parameter search, visualization, and the graphical interface are implemented in Python 3, using `psycopg2` for database access, `pandas` for data handling, and `matplotlib` together with `PyQt5` for plotting and the GUI itself. A full grid search over the smoothing parameters α, β, γ at a step size of 0.1 evaluates $9^3 = 729$ parameter combinations per drink and completes within a few seconds on this hardware.

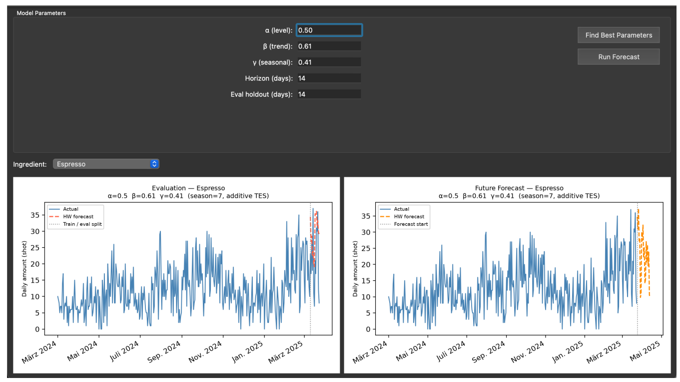

Fig. 1. The graphical interface showing model parameters, the ingredient, and the resulting evaluation and future forecast plots.

B. User Experience

Users can interact with the demo through a graphical interface (see Fig. 1). First, the user can select the parameter manually or run a grid search to find the best parameters for the forecasting model. Afterwards, the user can run the forecasting model and visualize the results for one ingredient. For the selected ingredient 2 plots are created, one comparing actual sales against the forecast for the held-out evaluation period, and one showing the forecast for the upcoming days beyond the available data. After the forecast is generated and without running the forecasting model again, the user can select another ingredient in a drop down menu and visualize the results for that ingredient. Visitors are also encouraged to compare ingredients that differ in how many drinks contribute to them. Ingredients drawn from several drinks tend to produce a stable forecast, since noise in any single drink's prediction is averaged out, whereas ingredients tied to only one or two drinks can inherit that drink's forecasting errors directly. This makes it possible to observe, within the same interface, both cases where the model performs convincingly and cases where its assumptions no longer hold.

C. Conclusion

The demo shows a complete path from raw, per-cup sales records to ingredient-level forecasts that could plausibly support purchasing decisions in a small coffee shop. Its particular value lies not in presenting a single polished result, but in letting visitors interactively expose the conditions under which a common forecasting method remains reliable and the conditions under which it does not, using the same pipeline and the same data throughout

References

References are excluded from the page limit, i.e., the bibliography might exceed the second page and it cannot be used to fill the 1.5 pages.